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Bi-Quasi-Arithmetic Mean Comparability

Vu Phan
Collin College

Abstract: In *Inequalities*, Hardy, Littlewood, and Polya addressed quasiarithmetic means (generalized means) and the comparison between any two of them. This paper presents the systematic combination of quasiarithmetic means leading to a larger concept: the biquasiarithmetic mean. Additionally, we state the conditions under which two biquasiarithmetic means are comparable. The article concludes with a glance at a further extension regarding multiquasiarithmetic means and their comparability.

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1. Introduction

Motivation

In 1934, Hardy, Littlewood, and Polya published *Inequalities*. In this classic treatise, they addressed many beautiful inequalities, such as

$$\text{Chebyshev's} \quad n\sum_{i=1}^{n} a_i b_i \geq \left(\sum_{i=1}^{n} a_i\right)\left(\sum_{i=1}^{n} b_i\right),$$

$$\text{Cauchy's} \quad \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \geq \left(\sum_{i=1}^{n} a_i b_i\right)^2, \text{ and}$$

$$\text{Minkowski's} \quad \sqrt[r]{\sum_{i=1}^{n} a_i^r} + \sqrt[r]{\sum_{i=1}^{n} b_i^r} \geq \sqrt[r]{\sum_{i=1}^{n} (a_i + b_i)^r}.$$

Beholding such aesthetic inequalities, a curious mind may wonder what they have in common. Hardy, Littlewood, and Polya had the answer: means. For instance, the aforementioned Chebyshev's Inequality can be rewritten as $\frac{\sum_{i=1}^{n} a_i b_i}{n} \geq \left(\frac{\sum_{i=1}^{n} a_i}{n}\right)\left(\frac{\sum_{i=1}^{n} b_i}{n}\right)$, in which three arithmetic means appear. In the same manner, Cauchy's Inequality may be rephrased as $\sqrt{\frac{\sum_{i=1}^{n} a_i^2}{n}}\sqrt{\frac{\sum_{i=1}^{n} b_i^2}{n}} \geq \frac{\sum_{i=1}^{n} a_i b_i}{n}$ to emphasize two quadratic means and an arithmetic one. Similarly, Minkowski's Inequality's alternative form $\sqrt[r]{\frac{\sum_{i=1}^{n} a_i^r}{n}} + \sqrt[r]{\frac{\sum_{i=1}^{n} b_i^r}{n}} \geq \sqrt[r]{\frac{\sum_{i=1}^{n} (a_i + b_i)^r}{n}}$ visualizes the power means.

Such means have several fascinating properties including symmetry, equivalency, and homogeneity. Nevertheless, the most interesting property is probably their comparability: which means are greater than others? This property is so compelling because of its direct relation to our primary interest: inequalities.

Past results

To study means, a promising strategy is to build up a generalized mean from fundamental ones. The exploration begins with the most basic mean: arithmetic $\frac{a+b}{2}$. Observing it beside quadratic $\sqrt{\frac{a^2+b^2}{2}}$ and harmonic $\frac{2}{\frac{1}{a}+\frac{1}{b}}$ means, we may notice the exponents and propose the generic power mean $\sqrt[r]{\frac{a^r+b^r}{2}}$. However, a significant difficulty emerges as the power mean seems to exclude the very basic geometric mean $\sqrt{ab}$. The problem remains stubbornly unsolved until we adventurously incorporate some calculus into the seemingly discrete field of inequalities: $\sqrt{ab}=\lim_{r\to 0}\sqrt[r]{\frac{a^r+b^r}{2}}$. This move creatively loosens the rigid definition of the power mean with the use of limits; nonetheless, on a larger scale, we are still confined within boundaries of the power function x^r. Will we achieve more by trying out other functions with the generic mean $f^{-1}\left(\frac{f(a)+f(b)}{2}\right)$? The answer is yes; for instance, we can now express the geometric mean without using explicit limits: $\sqrt{ab}=e^{\frac{\ln a+\ln b}{2}}$. This generalized mathematical object is formally known as the quasiarithmetic mean.

The discovery of quasiarithmetic means is a tremendous achievement; the exploration continues with the examination of the means' properties, especially their comparability: is $f^{-1}\left(\frac{f(a)+f(b)}{2}\right)$ greater than $g^{-1}\left(\frac{g(a)+g(b)}{2}\right)$? If two such means are comparable, it is due to some specific relation between the functions. By delicately maneuvering the already beautiful Jensen's Inequality $\sum_i p_i f(x_i)\geq f\left(\sum_i p_i x_i\right)$, Hardy, Littlewood, and Polya proved the relation between quasiarithmetic means' comparability and the functions' monotonicity and relative concavity. Quasiarithmetic means will be discussed in detail in Part 2.

New results

When dealing with inequalities, it is beneficial to focus on their quasiarithmetic means. Consider the following instance regarding a simple case of Minkowski's Inequality

$$\frac{\sqrt{\frac{a_1^2+a_2^2+a_3^2}{3}}+\sqrt{\frac{b_1^2+b_2^2+b_3^2}{3}}}{2} \geq \sqrt{\frac{\left(\frac{a_1+b_1}{2}\right)^2+\left(\frac{a_2+b_2}{2}\right)^2+\left(\frac{a_3+b_3}{2}\right)^2}{3}}.$$

Evidently, the left-hand side is a big arithmetic mean containing little quadratic means whereas the right-hand side is exactly the other way around. With M_a and M_q standing for arithmetic and quadratic means, the inequality can be rewritten as $M_a\left(M_q\left(a_1,a_2,a_3\right),M_q\left(b_1,b_2,b_3\right)\right) \geq M_q\left(M_a\left(a_1,b_1\right),M_a\left(a_2,b_2\right),M_a\left(a_3,b_3\right)\right)$.

Nevertheless, this notation fails to fully emphasize the most distinct characteristic of the initial inequality: the specific orientation in which points are distributed into means. This issue is fixed with new ideas depicted in the following illustration. First, let $X=\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$. Then, define a new concept: the biquasiarithmetic mean $M_{a,q}\left(X\right)=M_q\left(M_a\left(a_1,b_1\right),M_a\left(a_2,b_2\right),M_a\left(a_3,b_3\right)\right)$, which is exactly the inequality's right-hand side in this case. In a similar manner, we can express the left-hand side as another instance of the new mean and rephrase the inequality as $M_{q,a}\left(X^T\right) \geq M_{a,q}\left(X\right)$.

This expression clearly presents the most distinguished trait of Minkowski's Inequality. Recently, much effort was made to figure out whether the comparability between the generic $M_{g,f}\left(X^T\right)$ and $M_{f,g}\left(X\right)$ existed. However, it seems that the comparison is true only in the specific case $M_{q,a}\left(X^T\right) \geq M_{a,q}\left(X\right)$ of Minkowski's Inequality. Nonetheless, like quasiarithmetic means, biquasiarithmetic means are comparable in the traditional sense, namely $M_{f_1,f_2}\left(X\right) \geq M_{g_1,g_2}\left(X\right)$; the conditions predictably involve the functions' monotonicity and relative concavity. New findings on biquasiarithmetic means will be introduced in Part 3.

2. Quasiarithmetic Mean

Before examining these means in depth, we state the assumptions adopted throughout the paper.

Conventions 2.1

First, let $I \subseteq \mathsf{R}$ be an interval and $f,\ f_1,\ f_2,\ g,\ g_1, g_2 : I \to \square$ be continuous, strictly monotonic functions. We say x is a point if $x \in I$.

Next, $\mathbf{p} = (p_1, p_2, \ldots, p_n)$ is a tuple of weights if $p_1 + p_2 + \cdots + p_n = 1$ with $p_i \in \mathsf{R}^+$.

Then, when $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ and $\mathbf{z} = (z_1, z_2, \ldots, z_n)$, $\mathbf{y} \geq \mathbf{z}$ means $y_1 \geq z_1, y_2 \geq z_2, \ldots, y_n \geq z_n$.

The following definition introduces the quasiarithmetic mean, a generalized mean whose instances include quadratic, arithmetic, geometric, and harmonic means. This mean provides the basis for a larger concept, the biquasiarithmetic mean, which will be presented in Definition 3.1.

Definition 2.2: Quasiarithmetic Mean[1]

Let f be a function, $\mathbf{p} = (p_1, p_2, \ldots, p_n)$ be a tuple of weights, and $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ be a tuple of points. The corresponding quasiarithmetic mean is:

$$M_f(\mathbf{p}, \mathbf{x}) = f^{-1}\left(p_1 f(x_1) + p_2 f(x_2) + \cdots + p_n f(x_n)\right).$$

The next instance introduces some simple quasiarithmetic means.

Example 2.3

Let $f_1(x) = x^2$ and $g_1(x) = \ln x$. Take a tuple of weights $\mathbf{p}_1 = (\alpha, \beta)$. Next, let $\mathbf{x}_1 = (a_1, b_1)$, $\mathbf{x}_2 = (a_2, b_2)$, and $\mathbf{x}_3 = (a_3, b_3)$.

Based on Definition 2.2, the following are quadratic means: $M_{f_1}(\mathbf{p}_1, \mathbf{x}_1) = \sqrt{\alpha a_1^2 + \beta b_1^2}$, $M_{f_1}(\mathbf{p}_1, \mathbf{x}_2) = \sqrt{\alpha a_2^2 + \beta b_2^2}$, and $M_{f_1}(\mathbf{p}_1, \mathbf{x}_3) = \sqrt{\alpha a_3^2 + \beta b_3^2}$. These expressions are geometric means: $M_{g_1}(\mathbf{p}_1, \mathbf{x}_1) = a_1^{\alpha} b_1^{\beta}$, $M_{g_1}(\mathbf{p}_1, \mathbf{x}_2) = a_2^{\alpha} b_2^{\beta}$, and $M_{g_1}(\mathbf{p}_1, \mathbf{x}_3) = a_3^{\alpha} b_3^{\beta}$.

The next section formally proves an expected property of quasiarithmetic means: when all points increase or decrease, their quasiarithmetic means vary in the same direction. This fact will be used as a lemma in the proof of Theorem 3.3.

[1] This is presented in [1, p. 65, Definitions]. A clearer version is introduced in [2, pp. 2-3, Results for basic case].

Corollary 2.4: Monotonicity of the Quasiarithmetic Mean

Let f be a function, $\mathbf{p}=(p_1,p_2,\ldots,p_n)$ be a tuple of weights, and $\mathbf{y}=(y_1,y_2,\ldots,y_n)$ and $\mathbf{z}=(z_1,z_2,\ldots,z_n)$ be tuples of points. If $\mathbf{y}\geq\mathbf{z}$, then

$$M_f(\mathbf{p},\mathbf{y})\geq M_f(\mathbf{p},\mathbf{z}),$$

with equality if and only if $\mathbf{y}=\mathbf{z}$.

Proof

In Conventions 2.1, we assume that f is strictly monotonic, so f is either strictly increasing or strictly decreasing. Because the proofs are similar, it is sufficient to consider solely the case in which f is strictly decreasing.

Suppose $\mathbf{y}\geq\mathbf{z}$. Based on Conventions 2.1,

$$y_1\geq z_1,\ y_2\geq z_2,\ldots,y_n\geq z_n.$$

Applying the strictly decreasing function f to both sides of each of these inequalities, we get

$$f(y_1)\leq f(z_1),\ f(y_2)\leq f(z_2),\ldots,f(y_n)\leq f(z_n).$$

Recall that $p_i>0$ (Conventions 2.1). After multiplying both sides of each of these inequalities by its corresponding weight p_i and adding them together, we have

$$p_1f(y_1)+p_2f(y_2)+\cdots+p_nf(y_n)\leq p_1f(z_1)+p_2f(z_2)+\cdots+p_nf(z_n).$$

Because f is strictly decreasing, f^{-1} is strictly decreasing. Applying f^{-1} to both sides of the previous inequality, we have

$$f^{-1}\left(p_1f(y_1)+p_2f(y_2)+\cdots+p_nf(y_n)\right)\geq f^{-1}\left(p_1f(z_1)+p_2f(z_2)+\cdots+p_nf(z_n)\right).$$

In Definition 2.2, when choosing $\mathbf{x}$ to be $\mathbf{y}$, we get $M_f(\mathbf{p},\mathbf{y})=f^{-1}\left(p_1f(y_1)+p_2f(y_2)+\cdots+p_nf(y_n)\right)$, and when choosing $\mathbf{x}$ to be $\mathbf{z}$, we have $M_f(\mathbf{p},\mathbf{z})=f^{-1}\left(p_1f(z_1)+p_2f(z_2)+\cdots+p_nf(z_n)\right)$. Therefore, the latest inequality is indeed $M_f(\mathbf{p},\mathbf{y})\geq M_f(\mathbf{p},\mathbf{z})$, our desired result.

Finally, we examine when equality holds. If $\mathbf{y}=\mathbf{z}$, all the above inequalities become equalities, and $M_f(\mathbf{p},\mathbf{y})=M_f(\mathbf{p},\mathbf{z})$. If $\mathbf{y}\neq\mathbf{z}$, from the hypothesis $\mathbf{y}\geq\mathbf{z}$, we know that $y_i>z_i$ for some i. Evidently, certain inequalities above become strict, and $M_f(\mathbf{p},\mathbf{y})>M_f(\mathbf{p},\mathbf{z})$. Hence, equality holds if and only if $\mathbf{y}=\mathbf{z}$.

The next theorem states the sufficient condition for the comparability between two quasiarithmetic means. This proposition is crucial to the proof of Theorem 3.3: biquasiarithmetic mean comparability.

Theorem 2.5: Quasiarithmetic Mean Comparability [2]

Let f and g be functions, $\mathbf{p}=(p_1,p_2,\ldots,p_n)$ be a tuple of weights, and $\mathbf{x}=(x_1,x_2,\ldots,x_n)$ be a tuple of points. If either

i. f is strictly increasing and $f\circ g^{-1}$ is strictly concave upward, or

ii. f is strictly decreasing and $f\circ g^{-1}$ is strictly concave downward,

then

$$M_f(\mathbf{p},\mathbf{x})\geq M_g(\mathbf{p},\mathbf{x}),^{3}$$

with equality if and only if all points x_i are equal.[4]

3. Biquasiarithmetic Mean

More generalized than quasiarithmetic means is the biquasiarithmetic mean, which is defined with two functions, two tuples of weights, and a matrix of points. Basically, a biquasiarithmetic mean is generated by two turns of taking quasiarithmetic means. In the first turn, we take a quasiarithmetic mean of each column of the matrix of points; the yielded values are arranged as a row. In the second turn, we take the quasiarithmetic mean of that row, and the result is the biquasiarithmetic mean of the matrix of points.

Definition 3.1: Biquasiarithmetic Mean

Let f_1 and f_2 be functions and $\mathbf{p}_1=(p_{11},p_{12},\ldots,p_{1m})$ and $\mathbf{p}_2=(p_{21},p_{22},\ldots,p_{2n})$ be tuples of weights.

Next, let x_{ij} be points organized as $\mathbf{x}_1=\begin{bmatrix}x_{11}\\x_{21}\\\vdots\\x_{m1}\end{bmatrix}$, $\mathbf{x}_2=\begin{bmatrix}x_{12}\\x_{22}\\\vdots\\x_{m2}\end{bmatrix},\ldots,\mathbf{x}_n=\begin{bmatrix}x_{1n}\\x_{2n}\\\vdots\\x_{mn}\end{bmatrix}$ with $X=\begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_n\end{bmatrix}$.

Now, let $y_1=M_{f_1}(\mathbf{p}_1,\mathbf{x}_1)$, $y_2=M_{f_1}(\mathbf{p}_1,\mathbf{x}_2),\ldots,y_n=M_{f_1}(\mathbf{p}_1,\mathbf{x}_n)$ and $\mathbf{y}=\begin{bmatrix}y_1 & y_2 & \ldots & y_n\end{bmatrix}$. Then, the biquasiarithmetic mean corresponding to functions f_1 and f_2, tuples of weights $\mathbf{p}_1$ and $\mathbf{p}_2$, and points x_{ij} is:

$$M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)=M_{f_2}(\mathbf{p}_2,\mathbf{y}).$$

In later sections, we will frequently deal with another generic biquasiarithmetic mean corresponding to functions g_1 and g_2. That mean is specified through a similar procedure:

$z_1=M_{g_1}(\mathbf{p}_1,\mathbf{x}_1),z_2=M_{g_1}(\mathbf{p}_1,\mathbf{x}_2),\ldots,z_n=M_{g_1}(\mathbf{p}_1,\mathbf{x}_n)$, $\mathbf{z}=\begin{bmatrix}z_1 & z_2 & \ldots & z_n\end{bmatrix}$, *and*

$M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)=M_{g_2}(\mathbf{p}_2,\mathbf{z})$. *This mean-generating process is instanced below.*

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[2] The major claim is asserted in [1, p. 70, Theorem 85] and restated in [1, p. 75, Theorem 92]; the equality case is stated in [1, pp. 74-75, Theorem 90]. A more explicit statement is presented in [2, p. 3, Corollary 2.1].

[3] The proof for the general inequality case is given in [1, pp. 69-70, Comparability]; a more detailed version is found in [3, pp. 260-261, Lemma 1].

[4] The equality case is argued in [1, pp. 74-75, Equality in the fundamental inequalities].

Example 3.2

Reconsidering Example 2.3: $f_1(x)=x^2$, $g_1(x)=\ln x$, $\mathbf{p}_1=(\alpha,\beta)$, $\mathbf{x}_1=\begin{bmatrix}a_1\\b_1\end{bmatrix}$, $\mathbf{x}_2=\begin{bmatrix}a_2\\b_2\end{bmatrix}$, and $\mathbf{x}_3=\begin{bmatrix}a_3\\b_3\end{bmatrix}$. In addition, $M_{f_1}(\mathbf{p}_1,\mathbf{x}_1)=\sqrt{\alpha a_1^2+\beta b_1^2}$, $M_{f_1}(\mathbf{p}_1,\mathbf{x}_2)=\sqrt{\alpha a_2^2+\beta b_2^2}$, $M_{f_1}(\mathbf{p}_1,\mathbf{x}_3)=\sqrt{\alpha a_3^2+\beta b_3^2}$, $M_{g_1}(\mathbf{p}_1,\mathbf{x}_1)=a_1^\alpha b_1^\beta$, $M_{g_1}(\mathbf{p}_1,\mathbf{x}_2)=a_2^\alpha b_2^\beta$, and $M_{g_1}(\mathbf{p}_1,\mathbf{x}_3)=a_3^\alpha b_3^\beta$.

Now, let $f_2(x)=e^x$ and $g_2(x)=\sin x$. $\mathbf{p}_2=(\lambda_1,\lambda_2,\lambda_3)$ is another tuple of weights. Then, let $X=[\mathbf{x}_1\quad \mathbf{x}_2\quad \mathbf{x}_3]$.

Based on Definition 3.1,

$$\mathbf{y}=[y_1\quad y_2\quad y_3]=\left[M_{f_1}(\mathbf{p}_1,\mathbf{x}_1)\quad M_{f_1}(\mathbf{p}_1,\mathbf{x}_2)\quad M_{f_1}(\mathbf{p}_1,\mathbf{x}_3)\right]=\left[\sqrt{\alpha a_1^2+\beta b_1^2}\quad \sqrt{\alpha a_2^2+\beta b_2^2}\quad \sqrt{\alpha a_3^2+\beta b_3^2}\right].$$

Then, $M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)=M_{f_2}(\mathbf{p}_2,\mathbf{y})=\ln\left(\lambda_1 e^{\sqrt{\alpha a_1^2+\beta b_1^2}}+\lambda_2 e^{\sqrt{\alpha a_2^2+\beta b_2^2}}+\lambda_3 e^{\sqrt{\alpha a_3^2+\beta b_3^2}}\right)$.

In a similar manner, $\mathbf{z}=[z_1\quad z_2\quad z_3]=\left[M_{g_1}(\mathbf{p}_1,\mathbf{x}_1)\quad M_{g_1}(\mathbf{p}_1,\mathbf{x}_2)\quad M_{g_1}(\mathbf{p}_1,\mathbf{x}_3)\right]=\left[a_1^\alpha b_1^\beta\quad a_2^\alpha b_2^\beta\quad a_3^\alpha b_3^\beta\right]$ and $M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)=M_{g_2}(\mathbf{p}_2,\mathbf{z})=\sin^{-1}\left(\lambda_1\sin a_1^\alpha b_1^\beta+\lambda_2\sin a_2^\alpha b_2^\beta+\lambda_3\sin a_3^\alpha b_3^\beta\right)$.

Now, we can prove the most important result of the paper: the comparability between two biquasiarithmetic means.

Theorem 3.3: Biquasiarithmetic Mean Comparability

Let f_1, f_2, g_1, and g_2 be functions, $\mathbf{p}_1=(p_{11},p_{12},\ldots,p_{1m})$ and $\mathbf{p}_2=(p_{21},p_{22},\ldots,p_{2n})$ be tuples of weights, and x_{ij} be points organized as $\mathbf{x}_1=\begin{bmatrix}x_{11}\\x_{21}\\\vdots\\x_{m1}\end{bmatrix}$, $\mathbf{x}_2=\begin{bmatrix}x_{12}\\x_{22}\\\vdots\\x_{m2}\end{bmatrix},\ldots,\mathbf{x}_n=\begin{bmatrix}x_{1n}\\x_{2n}\\\vdots\\x_{mn}\end{bmatrix}$ with $X=[\mathbf{x}_1\quad \mathbf{x}_2\quad \ldots\quad \mathbf{x}_n]$.

If either
(i) f_1 is strictly increasing and $f_1\circ g_1^{-1}$ is strictly concave upward, or
(ii) f_1 is strictly decreasing and $f_1\circ g_1^{-1}$ is strictly concave downward,
and if either
(iii) f_2 is strictly increasing and $f_2\circ g_2^{-1}$ is strictly concave upward, or
(iv) f_2 is strictly decreasing and $f_2\circ g_2^{-1}$ is strictly concave downward,
then

$$M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)\geq M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X),$$

with equality if and only if all points x_{ij} are equal.

Proof

First, assume either

(i) f_1 is strictly increasing and $f_1 \circ g_1^{-1}$ is strictly concave upward, or

(ii) f_1 is strictly decreasing and $f_1 \circ g_1^{-1}$ is strictly concave downward.

Now, we apply Theorem 2.5, choosing $\mathbf{p}$ to be $\mathbf{p}_1$, f to be f_1, and g to be g_1. When choosing $\mathbf{x}$ to be $\mathbf{x}_1$, we get $M_{f_1}(\mathbf{p}_1, \mathbf{x}_1) \geq M_{g_1}(\mathbf{p}_1, \mathbf{x}_1)$. Similarly, when choosing $\mathbf{x}$ to be $\mathbf{x}_2$, we have $M_{f_1}(\mathbf{p}_1, \mathbf{x}_2) \geq M_{g_1}(\mathbf{p}_1, \mathbf{x}_2)$. In the same manner, we continue applying Theorem 2.5 with different $\mathbf{x}$ and get

$$M_{f_1}(\mathbf{p}_1, \mathbf{x}_1) \geq M_{g_1}(\mathbf{p}_1, \mathbf{x}_1), M_{f_1}(\mathbf{p}_1, \mathbf{x}_2) \geq M_{g_1}(\mathbf{p}_1, \mathbf{x}_2), \ldots, M_{f_1}(\mathbf{p}_1, \mathbf{x}_n) \geq M_{g_1}(\mathbf{p}_1, \mathbf{x}_n).$$

Recall from Definition 3.1 that $y_1 = M_{f_1}(\mathbf{p}_1, \mathbf{x}_1)$, $y_2 = M_{f_1}(\mathbf{p}_1, \mathbf{x}_2), \ldots, y_n = M_{f_1}(\mathbf{p}_1, \mathbf{x}_n)$ and $z_1 = M_{g_1}(\mathbf{p}_1, \mathbf{x}_1), z_2 = M_{g_1}(\mathbf{p}_1, \mathbf{x}_2), \ldots, z_n = M_{g_1}(\mathbf{p}_1, \mathbf{x}_n)$. Therefore, the above inequalities are actually

$$y_1 \geq z_1, y_2 \geq z_2, \ldots, y_n \geq z_n.$$

Additionally, according to Definition 3.1, $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ and $\mathbf{z} = (z_1, z_2, \ldots, z_n)$. Thus, based on Conventions 2.1, $\mathbf{y} \geq \mathbf{z}$.

Then, we apply Corollary 2.4, choosing $\mathbf{p}$ to be $\mathbf{p}_2$ and f to be f_2; the result is

(a) $M_{f_2}(\mathbf{p}_2, \mathbf{y}) \geq M_{f_2}(\mathbf{p}_2, \mathbf{z})$.

Now, assume either

(iii) f_2 is strictly increasing and $f_2 \circ g_2^{-1}$ is strictly concave upward, or

(iv) f_2 is strictly decreasing and $f_2 \circ g_2^{-1}$ is strictly concave downward.

We apply Theorem 2.5, choosing $\mathbf{p}$ to be $\mathbf{p}_2$, $\mathbf{x}$ to be $\mathbf{z}$, f to be f_2, and g to be g_2. The result is

(b) $M_{f_2}(\mathbf{p}_2, \mathbf{z}) \geq M_{g_2}(\mathbf{p}_2, \mathbf{z})$.

From results (a) and (b), $M_{f_2}(\mathbf{p}_2, \mathbf{y}) \geq M_{g_2}(\mathbf{p}_2, \mathbf{z})$. By Definition 3.1, $M_{f_1, f_2}(\mathbf{p}_1, \mathbf{p}_2, X) = M_{f_2}(\mathbf{p}_2, \mathbf{y})$ and $M_{g_1, g_2}(\mathbf{p}_1, \mathbf{p}_2, X) = M_{g_2}(\mathbf{p}_2, \mathbf{z})$. Thus, the most recent inequality is indeed $M_{f_1, f_2}(\mathbf{p}_1, \mathbf{p}_2, X) \geq M_{g_1, g_2}(\mathbf{p}_1, \mathbf{p}_2, X)$, our wanted result.

Finally, we examine when equality holds. If all points x_{ij} are equal, every previous inequality becomes an equality, and $M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)=M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)$. If the matrix X has at least two unequal elements, then either
(1) X has a column which contains unequal elements, or
(2) each column of X has its own equal elements; the unequal elements are in different columns.

In case (1), we can assume without loss of generality that the column $\mathbf{x}_1$ has unequal elements. With that hypothesis, $M_{f_1}(\mathbf{p}_1,\mathbf{x}_1)>M_{g_1}(\mathbf{p}_1,\mathbf{x}_1)$, which means $y_1>z_1$, so $\mathbf{y}\neq\mathbf{z}$, and $M_{f_2}(\mathbf{p}_2,\mathbf{y})>M_{f_2}(\mathbf{p}_2,\mathbf{z})$. Hence, $M_{f_2}(\mathbf{p}_2,\mathbf{y})>M_{g_2}(\mathbf{p}_2,\mathbf{z})$, meaning $M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)>M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)$.

In case (2), without loss of generality, we may hypothesize that the elements of columns $\mathbf{x}_1$ and $\mathbf{x}_2$ are respectively equal to distinct real numbers a_1 and a_2. Then, $z_1=M_{g_1}(\mathbf{p}_1,\mathbf{x}_1)=a_1$ and $z_2=M_{g_1}(\mathbf{p}_1,\mathbf{x}_2)=a_2$. Because $\mathbf{z}$ has unequal elements $z_1\neq z_2$, $M_{f_2}(\mathbf{p}_2,\mathbf{z})>M_{g_2}(\mathbf{p}_2,\mathbf{z})$. Consequently, $M_{f_2}(\mathbf{p}_2,\mathbf{y})>M_{g_2}(\mathbf{p}_2,\mathbf{z})$, which means $M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)>M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)$.

In summary, equality holds if and only if all points x_{ij} are equal.

The upcoming section illustrates an instance on applying biquasiarithmetic mean comparability.

Example 3.4

Recall Example 3.2: $f_1(x)=x^2$, $g_1(x)=\ln x$, $f_2(x)=e^x$, $g_2(x)=\sin x$, $\mathbf{p}_1=(\alpha,\beta)$, $\mathbf{p}_2=(\lambda_1,\lambda_2,\lambda_3)$, $\mathbf{x}_1=\begin{bmatrix}a_1\\b_1\end{bmatrix}$, $\mathbf{x}_2=\begin{bmatrix}a_2\\b_2\end{bmatrix}$, $\mathbf{x}_3=\begin{bmatrix}a_3\\b_3\end{bmatrix}$, and $X=\begin{bmatrix}\mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3\end{bmatrix}$. Additionally,
$M_{f_1,f_2}(\mathbf{p}_1,\mathbf{p}_2,X)=\ln\left(\lambda_1 e^{\sqrt{\alpha a_1^2+\beta b_1^2}}+\lambda_2 e^{\sqrt{\alpha a_2^2+\beta b_2^2}}+\lambda_3 e^{\sqrt{\alpha a_3^2+\beta b_3^2}}\right)$ and
$M_{g_1,g_2}(\mathbf{p}_1,\mathbf{p}_2,X)=\sin^{-1}\left(\lambda_1\sin a_1^{\alpha}b_1^{\beta}+\lambda_2\sin a_2^{\alpha}b_2^{\beta}+\lambda_3\sin a_3^{\alpha}b_3^{\beta}\right)$. Note that the common domain of f_1, g_1, f_2, and g_2 is $\left(0,\frac{\pi}{2}\right)$.

Now, we verify that the condition (i) of Theorem 3.3 is satisfied. Evidently, f_1 is strictly increasing because $f_1'(x)=2x>0$ on its domain $\left(0,\frac{\pi}{2}\right)$. Notice that $f_1\circ g_1^{-1}$ is strictly concave upward as $\left(f_1\circ g_1^{-1}\right)''(x)=4e^{2x}>0$.

After that, the condition (iii) is also confirmed. Obviously, f_2 is strictly increasing for $f_2'(x) = e^x > 0$. Also, $f_2 \circ g_2^{-1}$ is strictly concave upward since the derivative $\left(f_2 \circ g_2^{-1}\right)''(x) = \left(\frac{x}{\sqrt{1-x^2}} + 1\right)\frac{e^{\sin^{-1} x}}{1-x^2} > 0$ on its own domain $(0,1)$.

With appropriate hypotheses satisfied, we can conclude that $M_{f_1,f_2}(\mathbf{p}_1, \mathbf{p}_2, X) \geq M_{g_1,g_2}(\mathbf{p}_1, \mathbf{p}_2, X)$; explicitly,

$$\ln\left(\lambda_1 e^{\sqrt{\alpha a_1^2 + \beta b_1^2}} + \lambda_2 e^{\sqrt{\alpha a_2^2 + \beta b_2^2}} + \lambda_3 e^{\sqrt{\alpha a_3^2 + \beta b_3^2}}\right) \geq \sin^{-1}\left(\lambda_1 \sin a_1^{\alpha} b_1^{\beta} + \lambda_2 \sin a_2^{\alpha} b_2^{\beta} + \lambda_3 \sin a_3^{\alpha} b_3^{\beta}\right).$$

4. Conclusion

A journey initiated by the yearning for pure mathematics rarely has an expected destination. The new work presented in this paper began for the sole purpose of generalizing Minkowski's Inequality. That conjecture seemed to be incorrect, yet the exploration was continued. Recent achievements include findings of biquasiarithmetic means and their comparability.

Furthermore, a larger concept—the multiquasiarithmetic mean—has been proposed lately. Like the way biquasiarithmetic means were built, a multiquasiarithmetic mean is constructed by combining many quasiarithmetic means in specific orientations. It is defined with n functions, n tuples of weights, and points organized as an extended version of a matrix (in which every element has n indexes). Additionally, the comparability between any two multiquasiarithmetic means has been known to rely on their functions' monotonicity and relative concavity, like that of biquasiarithmetic means. These findings have been formally proved, but the current notations are excessively sophisticated, so further adjustment is needed before the work can be introduced.

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